ObsSec Recommendations for Lent Term 2026

Introduction

This document contains a selection of our recommended objects to look at with the telescopes this term. It is by no means a comprehensive list of everything there is to see, but instead contains some of the brighter or otherwise interesting objects, and provides a good starting point for visual observing. We've included no photos of anything, since these often make your own observations pale in comparison and distract you from appreciating that you can see the objects with your *very own eyes.*

The selection of objects is organised by constellation, which are themselves organised roughly in increasing Right Ascension (RA). As a result, objects listed earlier in the guide are highest in the sky and best seen earlier in the term or earlier in the evening. Objects near the end are best seen later in the term or later at night.

For each object, there is:

- The catalog name *[Messier (M), Index (IC), New General (NGC)]* and any common name.
- Object class, described below.
- Our difficulty rating, explained below.
- The apparent magnitude, which corresponds to how bright the object is. A smaller magnitude means brighter, and any object brighter than ~5 can be seen with the naked eye with our levels of light pollution.
- Celestial coordinates, in RA and Dec (Declination).
- A brief description of the object, with some observing tips and specific points of interest.

Difficulty scale:

- Easy
  These objects are great to start with. They will typically be easy to find, have lots of visible features, and will jump out of the eyepiece at you.
- Medium
  These are harder to spot, requiring some effort to find and especially to make out detail. Moonless nights and turning off the dome lights will help.
- Hard
  Most of these objects need some experience to observe, including long dark adaptation and averted vision. Detail can be very tricky to discern, and often a successful observation reveals just a faint blur and the knowledge that you've seen something really exciting with your own eyes.

- Insane or Astrophotography
  These objects might well be impossible to see in Cambridge, requiring larger telescope apertures and darker skies than we have here. Do give them a go, however, and many of them are also great targets for astrophotography if you have the ability to do that.

Object Classes:

- **Planets**
  Large bodies orbiting our sun. You are standing on one of them! Through a telescope, they are small and bright, so take magnification (via smaller eyepieces) well. Details on their surfaces are limited by seeing (atmospheric turbulence) instead of contrast (from light pollution).
- **Multiple star systems**
  Stars that are physically close to and orbit each other. The contrast between different star types is readily apparent for closely separated ones. Some particularly tight pairs can require good seeing to 'split' (i.e. to convince yourself that the point of light is actually two stars).
- **HII regions**
  Large glowing emission nebulae, primarily of hydrogen gas. These act as stellar nurseries, and the newly formed stars within them ionise the gas to glow a deep red, among other colours (which are usually only visible in photos). Often only unusually bright regions are visible visually, although filters can help increase the contrast.
- **Planetary nebulae**
  Named for being small, round, relatively bright, and therefore planet-like, these emission nebulae are glowing shells of gas surrounding hot dying stars. Their small size and higher contrast make them great targets for teasing out difficult detail in a large focal length telescope like the Northumberland.
- **Supernova remnants**
  Emission nebulae formed from the violent explosion of the progenitor star at the end of its life. While rare and (astronomically) short-lived, we do have the *Crab Nebula* (M1) visible with the telescopes, appearing similar to a faint planetary nebula.
- **Reflection nebulae**
  Often combined with other nebula types, these are simply dust clouds reflecting broadband starlight. As a result, they appear white in photographs and are not improved with filters, sometimes necessitating darker skies than we have here.
- **Open clusters**
  Groups of stars which all formed together, and younger ones are often associated with the HII region that birthed them. They appear as gatherings of tens to hundreds of easily distinguishable points.
- **Globular clusters**
  Collections of thousands to hundreds of thousands of gravitationally bound stars. These are often old and distant, outside the plane of our galaxy. They can be seen as hazy globes, in which you can often discern the faint speckling of the thousands of constituent stars.
- **Galaxies**
  These are the most distant objects easily seen, and are collections of tens of billions of stars and all the other objects discussed here. They are usually visible through the telescopes as elliptical clouds with a bright core, and some have subtle detail from spiral arms or dust lanes that can be picked out with practice.

Good luck observing, and clear skies!

- *The ObsSecs, 2026*

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The Lent Term sky, looking southwards. Image taken from Stellarium. (Planet Jupiter in Gemini as an unlabeled bright orb)

Aries and Pisces

M74 - Phantom Galaxy Hard (Mag 9.39)

RA 1h 36m 42s | Dec +15° 47′ 1″

M74 is an edge-on spiral galaxy in the constellation *Pisces*, 32 million light-years away from Earth. The galaxy contains two clearly defined spiral arms and is therefore used as an archetypal example of a grand design spiral galaxy. Its low surface brightness makes it the most difficult Messier object for amateur astronomers to observe, and is a good challenge.

NGC 772 - Fiddlehead Galaxy Insane or Astrophotography (Mag 11.10)

RA 1h 59m 19s | Dec +19° 00′ 27″

NGC 772 is a stunning early spiral system in the constellation *Aries*, seen 37° from edge-on. It has several tightly coiled arms and one strong, asymmetric arm crossing the disk westwards. The one bright arm coils around the nucleus like a fiddlehead unfolding.

NGC 877 (Galaxy) Insane or Astrophotography (Mag 11.89)

RA 2h 18m 00s | Dec +14° 32′ 38″

NGC 877, a spiral galaxy in the constellation *Aries*, features two spiral arms with a grand design pattern and slightly disturbed morphology. When pictured in H-alpha, the arms have numerous knots and appear brighter than the nucleus.

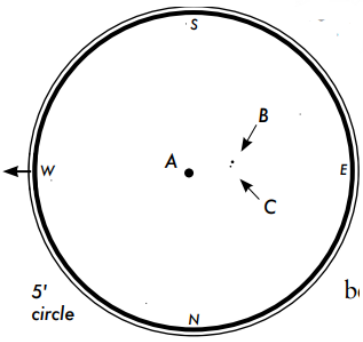
Eridanus and Cetus

Keid (Multiple Star) Medium (Mag 4.42 & 9.72)

RA 4h 15m 16s | Dec −07° 39′ 10″ (Keid A)

Keid (Omicron[2] Eridani) is a system of 3 stars in *Eridanus*, visible to a telescope, and the easiest main sequence & white dwarf binary system to see in the sky. High magnification is recommended.

Keid A is a K-type star, a bit fainter than our Sun, located only 16.5 light years from us. Its distant companions B and C orbit more than 400 AU from A, circling each other at an average distance of about 35 AU.



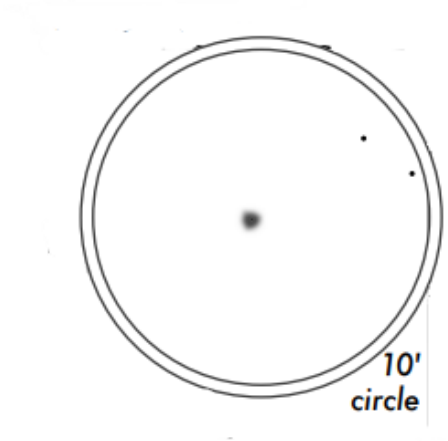
Keid B is a white dwarf, with half the mass of our Sun packed into a volume no bigger than the Earth. Keid B is the easiest white dwarf to see, since the other nearby white dwarfs, the companions of *Procyon* and *Sirius*, are very close to very bright stars. Keid C is a small red M-type star. Though the smallest of these stars, it burns the slowest and so will continue to shine long after Keid A and Keid B have turned to embers.

Photo credit: "Turn Left at Orion" - Guy Consolmagno, Dan M. Davis)

NGC 1535 - Cleopatra's Eye (Planetary Nebula) Medium (Mag 9.39)

RA 4h 14m 15s | Dec −12° 44′ 21″

Cleopatra's Eye is a planetary nebula in *Eridanus*, that is, the ball of gas erupted from a dying sun-like star, leaving behind a central white dwarf.



At low power you can spot the planetary nebula as a small but fuzzy blue star.

Once you find it, switch to your highest power to see it as a dreamy blue disk. Very high power and larger apertures reveal a lumpy, mottled surface; under excellent conditions you may see the 11th-magnitude central star.

Photo credit: "Turn Left at Orion" - Guy Consolmagno, Dan M. Davis)

M77 - Squid Galaxy Medium (Mag 8.89)

RA 2h 42m 41s | Dec -0° 0′ 48″

The Squid Galaxy is a barred spiral galaxy in *Cetus*, with an active radio nucleus, and is the prototype of the peculiar Seyfert galaxies - galaxies with quasar-like nuclei and high surface brightness.

Perseus

NGC 869 & 884 - Perseus Double Cluster Easy

RA 2h 20m | Dec +57° 8′

The Double Cluster in *Perseus* consists of the open clusters NGC 869 and NGC 884, which lie at a distance of about 7,500 light-years in the Perseus Arm of the Milky Way galaxy. It has a very high declination of +57 making it very inconvenient for the Northumberland Telescope, but can be observed with the Dobsonian or Thorrowgood telescopes much more easily.

NGC 869 has a mass of 4,700 solar masses and NGC 884 weighs in at 3,700 solar masses; both clusters are surrounded with a very extensive halo of stars, with a total mass for the complex of at least 20,000 solar masses. They form the core of the Perseus OB1 association of young hot stars. Based on their individual stars, the clusters are relatively young, both 14 million years old. In comparison, the *Pleiades* have an estimated age of 100 million years.

M34 - Spiral Cluster Easy

RA 2h 42.1m | Dec +42° 46′

M34 is a large and relatively near open cluster in *Perseus*, with apparent dimensions of 35'. It is about 1,500 light years away, and its age lies between the ages of the *Pleiades* open cluster at 100 million years and the *Hyades* open cluster at 800 million years.

NGC 1499 - California Nebula (HII Region) Insane or Astrophotography

RA 4h 3m 18s | Dec +36° 25′ 18″

The California Nebula is an emission nebula 1,300 light years away in *Perseus*. It's almost 2.5° long in the sky and because of its very low surface brightness, it is extremely difficult to observe visually, and is recommended for astrophotographers only.

Orion

M42 - The Orion Nebula (HII Region) Easy (Mag 4.00)

RA 5h 35m 17s | Dec -5° 23′ 28″

The Orion Nebula is a diffuse, star forming nebula with a prominent HII region, situated south of *Orion's Belt* and is known as the middle "star" in the "sword" of *Orion*. It is one of the brightest nebulae and is visible to the naked eye in the night sky with an apparent magnitude of 4.0. It is 1,350 light-years away and is the closest region of massive star formation to Earth. This is something not to be missed!

The nebula contains a very young, characteristic open cluster, known as the *Trapezium Cluster* due to the asterism of its primary four stars within a diameter of 1.5 light years. Two of these can be resolved into their component binary systems on nights with good seeing, giving a total of six stars. This trapezium cluster lights up the nebula.

M43 - De Mairan's Nebula (HII Region) Medium (Mag 9.00)

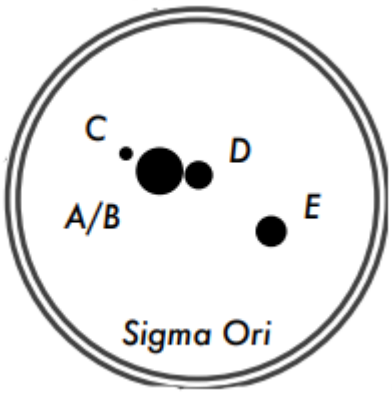
RA 5h 35m 32s | Dec −05° 17′ 57″

M43 or De Mairan's Nebula is a star-forming nebula with a prominent HII region, physically part of the *Orion Nebula (M42)*, but separated from the main nebula by a dense lane of dust known as the northeast dark lane. It is part of the much larger Orion Molecular Cloud complex.

Sigma Orionis (Multiple Star) Easy (Mag 3.79, 8.43)

RA 5h 38m 42s | Dec −02° 36′ 00″

Sigma Orionis is a multiple star system in the constellation *Orion*, consisting of the brightest members of a young open cluster. It is found at the eastern end of the belt, south west of *Alnitak* and west of the *Horsehead Nebula* which it partially illuminates. High magnification is recommended.



A and B are a pair of extremely bright and massive stars, only about 100 AU apart. Both C and D are separated from A by great distances, at least 3,800 AU and 4,500 AU respectively, and E is almost a quarter of a light year from A.

Photo credit: "Turn Left at Orion" - Guy Consolmagno, Dan M. Davis)

M78 - Ghost Nebula: (Reflection Nebula & HII Region) Hard (Magnitude 8.30)

RA 5h 46m 47s | Dec +0° 0′ 50″

M78 or the Ghost Nebula is the brightest diffuse reflection nebula in a group that includes NGC 2064, NGC 2067, and NGC 2071, all part of the Orion B molecular cloud complex, in the constellation *Orion*.

Located approximately 1,350 light-years from Earth, M78 is visible in small telescopes as a hazy patch illuminated by two B-type stars, of 10th and 11th magnitude.
The nebula's star-forming dust cloud reflects light from its two central stars, making it visible.

(Photo Credit: Deep Sky Watch)

NGC 2024 - Flame Nebula: (HII Region) Insane or Astrophotography (Magnitude 10.0)

RA 5h 41m 54s | Dec −01° 51′ 0″

The Flame Nebula, designated as NGC 2024 and Sh2-277, is an emission nebula in the constellation *Orion*. It is about 1350 light-years away. At that distance, the Flame Nebula lies within the Orion B cloud of the larger Orion Molecular Cloud Complex, near the star *Alnitak* in *Orion's belt* (not physically associated with the star).

Taurus

Uranus Easy (Magnitude 5.70)

The planet Uranus is visible as a small blue disc in the constellation of *Taurus* in Lent 2026, near the *Pleiades* star cluster. The 14th magnitude moons *Titania* and *Oberon* can be seen with high magnification when conditions are excellent.

M45 - Pleiades (Open Cluster) Easy

RA 3h 47m | Dec +24° 7′

The Pleiades is an open star cluster containing young B-type stars in the northwest of the constellation *Taurus*. At a distance of about 444 light-years, it is among the nearest star clusters to Earth and the nearest Messier object to Earth, being the most obvious star cluster to the naked eye in the night sky. Reflection nebulae around the brightest stars were once thought to be leftover material from their formation, but are now considered likely to be an unrelated dust cloud in the interstellar medium through which the stars are currently passing.

The Pleiades is best observed by binoculars, rather than a high magnification refractor such as the Northumberland or Thorrowgood telescopes. For an added challenge, you can look for the reflection nebulosity that surrounds the main stars in the cluster.

Hyades (Open Cluster) Easy

RA 4h 27m | Dec +15° 52′

The Hyades is the nearest open star cluster to the Earth, and is 153 light years away, and its age is estimated to be about 625 million years. It is located in the constellation *Taurus*.

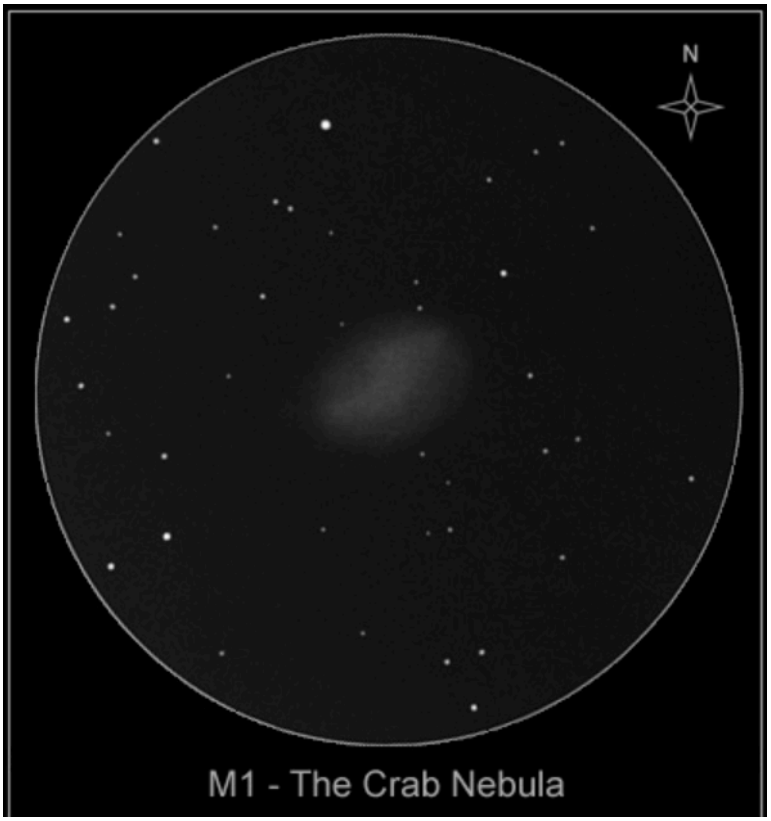
The five brightest member stars of the Hyades have consumed the hydrogen fuel at their cores and have evolved into giant stars. Four of these stars, with Bayer designations Gamma, Delta1, Epsilon, and Theta2 Tauri, form an asterism that is traditionally identified as the head of Taurus the Bull. The fifth of these stars is Theta1 Tauri, a tight naked-eye companion to the brighter Theta2 Tauri. Epsilon Tauri, known as Ain (the "Bull's Eye"), has a gas giant exoplanet candidate, the first planet to be found in any open cluster.

The Hyades is also best observed by binoculars, rather than a high magnification refractor such as the Northumberland or Thorrowgood telescopes.

M1 - Crab Nebula (Supernova Remnant) Hard (Mag 8.39)

RA 5h 34m 32s | Dec +22° 0′ 52″

The Crab Nebula is a supernova remnant that is faintly visible using our telescopes on nights with good transparency, located in the constellation *Taurus*. It corresponds with a bright supernova observed in 1054 CE by Mayan, Japanese, and Arab stargazers; this supernova was also recorded by Chinese astronomers as a guest star. The nebula was the first astronomical object identified that corresponds with a historically-observed supernova explosion.



(Photo Credit: Deep Sky Watch)

Canis Major, Puppis & Lepus

M46 (Open Cluster): Easy (Mag 6.09)

RA 7h 41m 46s | Dec −14° 48′ 36″

M46 is an open cluster of stars in the slightly southern constellation of *Puppis*. John Dreyer, publisher of the Index Catalogue (IC), described it as "very bright, very rich, very large." It is about 5,000 light-years away. There are an estimated 500 stars in the cluster with a combined mass of 453 solar masses, and it is thought to be 250 million years old.

The *planetary nebula NGC 2438* appears to lie within the cluster near its northern edge, but it is most likely unrelated since it does not share the cluster's radial velocity. For an added challenge, you may look for it - it appears as an out of focus star in the noted area.

M47 (Open Cluster): Easy (Mag 4.40)

RA 7h 36.6m | Dec -14° 30′

M47 is a coarse, bright, open cluster in the slightly southern constellation of *Puppis*. M47 is centered about 1,600 light-years away and is about 78 million years old. The cluster is dominated by hot class B main sequence and giant stars, but a noticeable colour contrast comes from its brightest red giants. It is about a degree from *Messier 46*, which is much older and much further away.

M41 (Open Cluster): Medium (Mag 4.50)

RA 6h 46m | Dec −20° 46′

M41 is a large, bright, open cluster in the southern constellation of *Canis Major*. The cluster spans an area comparable to the size of the full moon and contains about 100 stars, including several red giants and white dwarfs. The difficulty arises in distinguishing it from the City of Cambridge's light pollution, since it is close to the horizon and appears in the south.

IC 418 - Spirograph Nebula (Planetary Nebula) Medium (Mag 9.30)

RA 5h 27m 28s | Dec −12° 41′ 50

The Spirograph Nebula is a bright planetary nebula located in the constellation of *Lepus* about 3,600 ly away from Earth. It spans 0.3 light-years across. The 10th magnitude central star of the planetary nebula, HD 35914, is an O-type star with a spectral type of O7fp. The nebula formed a few thousand years ago during the star's last stages of its red giant phase, when material from the star's outer layers was ejected from the star into the surrounding space. The nebula's glow is caused by the central star's ultraviolet radiation interacting with the gas.

Sirius: (Multiple Star) Hard (Mag -1.46 & +8.50)

RA 6h 45m 09s | Dec −16° 42′ 58″

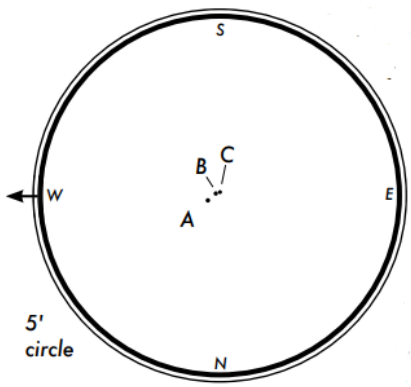
Sirius is the brightest star in the night sky, its brightness being primarily due to its close proximity, being 8.7 light years away, located in the constellation *Canis Major*.

It is a binary star consisting of a main-sequence star of spectral type A1V, termed Sirius A, and a faint *white dwarf* companion of spectral type DA2, termed Sirius B, the first white dwarf ever discovered. The distance between the two varies between 8.2 and 31.5 astronomical units as they orbit every 50 years, with Sirius B being as far away as possible from A this current year. (11.3 arc seconds separation)
To be able to identify the 8th magnitude companion, you should use a hexagonal mask, and be particularly patient.

Monoceros

M50 - Heart-Shaped Cluster Easy (Mag 5.90)

RA 7h 02m 47s | Dec −08° 20′ 16″
The Heart-shaped Cluster is an open cluster of stars in the constellation *Monoceros*. M50 is about 2,900 light-years away from Earth. It has a core radius of 5.9 ly (1.8 pc) and spans 17.8 ly (5.46 pc). The cluster has 508 confirmed and 109 probable members – their combined mass is more than 285 solar masses, the mean stellar density would thus be 1.3 stars per cubic parsec. It is around 140 million years old, with two high-mass white dwarfs and two chemically peculiar stars.
Traditionally considered to be a single star cluster, in 2025 it was found to consist of two separate sub-clusters (NGC 2323-a and NGC 2323-b), making it a binary cluster.



Beta Monocerotis (Multiple Star) Medium (Mag 4.6, 5.4, 5.6)

RA 6h 28m 49s | Dec −07° 01′ 59″
Beta Monocerotis is a beautiful triple star system in the constellation of *Monoceros*, about 700 light years from us, with special distinction being that they are all of a similar class. The three stars lie approximately in a straight line. Component B is 7" from component A, and component C a further 3" away. The white star, A, is about 1,500 AU from B. Stars B and C are a bit over 600 AU from each other. Given their great distances apart, they move very slowly about each other, taking thousands of years to complete one orbit.

(Photo Credit: "Turn Left at Orion" - Guy Consolmagno, Dan M. Davis)

NGC 2244 - Rosette Nebula: Insane or Astrophotography (Magnitude of Cluster +4.80)

RA 6h 31m 54s | Dec +4° 56′
The Rosette Nebula is a large, circular emission nebula (HII Region) in the constellation *Monoceros*, which surrounds an open cluster of hot, young stars known as the Rosette Cluster. The nebula is an HII region at a distance of 5,200 light-years, near one end of a large molecular cloud in *Monoceros*. Its diameter is about 130 light-years and the central hole is of 30 light years diameter, and the nebula contains 10,000 solar masses.

The cluster is closely associated with the nebulosity, having recently formed from the nebula itself. UV radiation from its hot O-type stars energizes the nebula, causing it to fluoresce, and generate large amounts of X-ray emission. The hollow in the centre has been blown out by stellar winds from this cluster of stars, and also leads to star formation by compressing the interstellar cloud.

The nebula is a vast cloud of dust and gas extending over a degree, and covers an area about 5 times that of the full moon, and is a target for astrophotographers. It may be seen visually only in very dark conditions with low magnification, and its distinctive red colour is only visible under photography.

Gemini

Jupiter Easy (Mag -2.50)

The planet Jupiter is visible in the constellation of *Gemini*. Currently an excellent time to view! Jupiter recently passed closest to the Earth (*Opposition*) on Jan 10, making this period the time when it looms largest and brightest in Earth's sky. Because Jupiter spins so quickly, its disk is visibly flattened. It is wider at the equator and narrower at the poles.

You should be able to pick out some of Jupiter's clouds, alternating light and dark. Zones are white, and belts are dark. Within the belts, on sharp nights of good seeing (correlated with low twinkling of stars) look for delicate and detailed wisps, and irregularities – *festoons* – that can be seen to move across the disk during the night as the planet spins.

Look for standout features like the *Great Red Spot* in Jupiter's southern hemisphere. Of course it is only visible when its side of Jupiter is rotated towards the Earth; and more than about an hour before or after its transit it's harder to see. Also, the 'red' aspect of the spot varies a lot over time.

Among Jupiter's **moons**, the four major moons (known as the Galilean Moons, after their discoverer, Galileo) are visible through our telescopes:
Io is bright, almost as bright as *Ganymede*, and slightly yellowish-orange.
Europa is more of a pure white, but dimmer.
Ganymede is the brightest moon.
Callisto is the darkest; often (though not always) it's seen the farthest from Jupiter.
At least three of Jupiter's moons are almost always visible at any given moment.

Because Jupiter's spin is only slightly tilted, and the orbits of the moons follow the spin, we always see the orbits of these moons more-or-less edge-on. From the eastern side of Jupiter, each crosses in front of the disk, continues to its maximum distance west of the planet, then moves back behind the disk to the eastern side again. The exception is *Callisto*: for a bit over half of its orbits it can be seen passing above or below Jupiter.

When a moon crosses in front of Jupiter so that its shadow, roughly the size of the moon itself, can fall on Jupiter (called a **Transit**) – can be seen on a good crisp night even in a small telescope. Sometimes during transits it is still possible to see the transiting

moon itself. Europa, which is white in color, stands out most easily against darker belts; dark *Callisto* can be seen against the brighter zones. It takes from two hours (for *Io*) to five hours (for *Callisto*) for a moon or its shadow to cross Jupiter's disk. The location of the transits moves north to south of Jupiter's equator depending on what part of its 12 year orbit we're observing. They will happen in Jupiter's southern hemisphere until late 2026.

For details on Galilean Moon transit timings or other such events on Jupiter, refer to the internet or a sky map app such as *Stellarium* or *SkySafari*, under Jupiter.

(Acknowledgement to Liu Hao, hl731@cam.ac.uk, & "Turn Left at Orion" - Guy Consolmagno, Dan M. Davis)

M35 - Shoe-Buckle Cluster Easy (Mag 5.09)

RA 6h 08m 54s | Dec +24° 20′ 00″

The Shoe-Buckle Cluster, is a relatively close open cluster of stars in the west of the constellation *Gemini*. It is scattered over part of the sky almost the size of the full moon and is 2,970 light-years away, with a linear diameter of about 24 light-years. It contains over 120 stars from 6th to 13th magnitude in a 30 arc minute area, and may contain more than 500 stars in total.

30 arc minutes away is the fainter but tighter *open cluster NGC 2158*, which appears in the same field of view when seen at low powers.

NGC 2392 - Eskimo or Clown Face Nebula (Planetary Nebula) Medium (Mag 9.19)

RA 7h 29m 11s | Dec +20° 54′ 42″

The Eskimo Nebula is a bipolar double-shell planetary nebula in the constellation of *Gemini*, 6,500 light years away. The formation resembles a person's head surrounded by a parka hood. It is surrounded by gas that composed the outer layers of a Sun-like star. The visible inner filaments are ejected by a strong wind of particles from the central star. The outer disk contains unusual, light-year-long filaments.

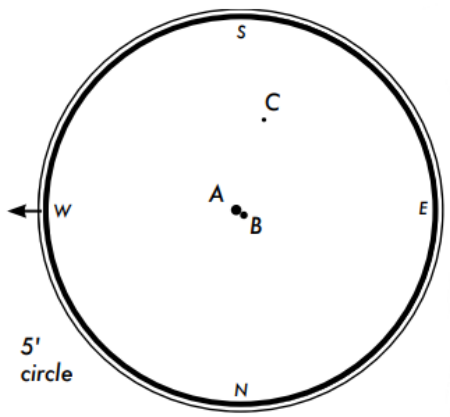
The cloud itself is about 40,000 AU across (about two thirds of a light year), and judging from the motions of the gas it appears to be growing at more than 20 AU per year. From this, one can infer that this nebula may be less than 2,000 years old, making it one of the youngest of the planetary nebulae.

Castor (Multiple Star) Medium (Mag 1.57, 2.97)

RA 7h 34m 36s | Dec +31° 53′ 18″

Castor (Alpha Geminorum), the second brightest star of the constellation *Gemini*, is a multiple star system made up of six individual stars; there are three visual components, all of which are spectroscopic binaries.

With good magnification and a clear night, you will see a definite separation into two fairly bright stars; then note a third, dimmer star with a slightly orange tint to the south-southeast.



A and B are each a pair of hot young A-type stars, roughly twice the size of our Sun and perhaps ten times as bright; star C is actually two dwarf K-type stars, about two thirds the size of our Sun and less than 3% as bright.

Each pair is in itself separated by only a few million miles; the A pair orbits once every nine days, B once every three days, and C in less than a day. The B pair orbits the A pair with a period of about 450 years. At their closest, they are only 1.8 arc seconds apart; at present, their separation is about four arc seconds, and growing slowly.

(Photo credit: "Turn Left at Orion" - Guy Consolmagno, Dan M. Davis)

Over the next hundred years star B will eventually move northwards, reaching a maximum separation of 6.5 arc seconds before circling back in its orbit towards A. By the middle of the twenty-first century, this will be an easy pair to split even in the smallest telescope. The C pair must be more than 1,000 AU from the other stars. They take over 10,000 years to complete an orbit around A and B.

Cancer

M44 - Beehive Cluster Easy (Mag 3.09)

RA 8h 40m 24s | Dec +19° 59′ 0″

The Beehive Cluster is an open cluster in the constellation *Cancer*. One of the nearest open clusters to Earth, at approximately 600 light-years away, it contains a larger population of stars than other nearby bright open clusters, holding around 1,000 stars, appearing 70 arc minutes in size. The cluster is thought to be quite young, at 600-700 million years old. Its age and proper motion coincide with those of the *Hyades*, suggesting they may share similar origins.

The Beehive Cluster is best observed by binoculars, rather than a high magnification refractor such as the Northumberland or Thorrowgood telescopes.

M67 - King Cobra Cluster Easy (Mag 6.90)

RA 8h 51.3m | Dec +11° 49′

The King Cobra Cluster is an open cluster in the southern, equatorial half of the constellation *Cancer*, and is one of the most ancient open clusters known. Estimates of its age range between 3.2 and 5 billion years. Distance estimates are likewise varied, but typically are 800–900 parsecs (2,600–2,900 ly). It has more than 100 stars similar to the Sun, and numerous red giants. The total star count has been estimated at well over 500.

The cluster contains no main sequence stars bluer (hotter) than spectral type F, other than perhaps some of the blue stragglers, since the brighter stars of that age have already left the main sequence. It appears that M67 has a bias toward heavier stars. One cause of this is mass segregation, the process by which lighter stars gain speed at the expense of more massive stars during close encounters, which moves them to greater average distance from the center of the cluster or allows escape altogether.

Auriga

M36, M37 and M38 form a trio of brilliant open clusters in the constellation *Auriga*.

M36 - Pinwheel Cluster Easy (Mag 6.00)

RA 5h 36m 18s | Dec +34° 08′ 24″

M36 is an open cluster of 60 young stars, about 15 light years in diameter and located 4,000 light years away from us, in the constellation of *Auriga*. The brightest of the stars are several hundred times as bright as our Sun. They are mostly type-B stars – bright, blue, and hot. With its pretty knot of jewel-like blue stars against the background of hazy light from the dozens of stars too faint to be made out individually in a small telescope, this is an easy and very rewarding object. It is most impressive to look at under low power. That way you get both the individual bright stars and the grainy haze of unresolved stars around them.

M37 - Salt and Pepper Cluster Easy (Mag 5.59)

RA 5h 52m 18s | Dec +32° 33′ 02″

M37 is the brightest and richest open cluster in the constellation *Auriga*, and not to be missed. This cluster is made up of several hundred stars, with 150 of them just bright enough (11th to 13th magnitude) to be picked out easily. It exists in the antipodal direction, opposite from the Galactic Center as seen from Earth, so is in one of the nearby outer arms, 4,500 light-years away from us.

M38 - Starfish Cluster Easy (Mag 5.59)

RA 5h 28m 43s | Dec +35° 51′ 18″

M38 is a neighbor of the other two open clusters M36 and M37, located about 3,500 light years away from us, in the constellation *Auriga*. With about 100 stars in a cloud about 20 light years across, it is in between M36 and M37 in size.

IC 405 - Flaming Star Nebula (HII Region) Insane or Astrophotography (Mag 10.00)

RA 5h 16m 5s | Dec +34° 27′ 49″

The Flaming Star Nebula is an emission and reflection nebula in the constellation *Auriga* surrounding the bluish, irregular variable star *AE Aurigae*. The nebula measures approximately 37.0' x 19.0', lies about 1,500 light-years away from Earth, and about 5 light-years across. It is believed that the proper motion of the central star can be traced back to the *Orion's belt* area.

Lynx

NGC 2419 - Intergalactic Wanderer (Globular Cluster) Medium (Mag 10.30)

RA 7h 38m 08s | Dec +38° 52′ 55″

NGC 2419 is a globular cluster in the constellation *Lynx*. It bears the nickname "the Intergalactic Wanderer," which was bestowed when it was erroneously thought not to be in orbit around the Milky Way. Its orbit takes it farther away from the galactic center than the Magellanic Clouds, but it can (with qualifications) be considered as part of the Milky Way. At this great distance it takes three billion years to make one trip around the galaxy.

Intrinsically it is one of the brightest and most massive globular clusters of our galaxy, having an absolute magnitude of −9.42 and being 900,000 times more massive than the Sun. It was proposed that NGC 2419 could be, as *Omega Centauri*, which is in the southern constellation of Centaurus, the remnant of a dwarf spheroidal galaxy disrupted and accreted by the Milky Way.

Leo

M65, M66, NGC 3628 - Leo Triplet (Galaxies) Medium (Mag 9.3, 8.9, 9.5)

RA 11h 17m | Dec +13° 25′

The Spiral Galaxies M65, M66 & NGC 3628 form a group called the *Leo Triplet* or M66 group, in the constellation *Leo*.

M65 is an intermediate (in between barred & unbarred) spiral galaxy 35 million light-years away, observed from an oblique angle.

M66 is an intermediate spiral galaxy lying 31 million light-years away and is about 95 thousand light-years across. It has a morphological classification of SABb, indicating a spiral shape with a weak bar feature and loosely wound arms, which may be visible under photography. Gravitational interaction from its past encounter with neighboring NGC 3628 800 million years ago has resulted in an extremely high central mass concentration.

NGC 3628 is an unbarred spiral galaxy about 35 million light-years away, with relatively low surface brightness as compared to M65 & M66, making it hard to see in a telescope. Its most conspicuous feature is the broad and obscuring band of dust located along the outer edge of its spiral arms, effectively transecting the galaxy to the view from Earth.

M95, M96, M105 - Leo I Group (Galaxies) Hard (Mag 9.7, 9.3, 9.3)

RA 10h 48m | Dec+12° 40′

The Spiral Galaxies M95, M96, M105 form part of the Leo I or M96 Group of galaxies in the constellation *Leo*, which forms part of the *Virgo Supercluster*.

M95 is a barred spiral galaxy about 33 million light-years away.

M96 is an intermediate spiral galaxy about 31 million light-years away. It is categorized as a double-barred spiral galaxy with a small inner bulge through the core along with an outer bulge, which may be seen under photography. It is a very asymmetric galaxy; its dust and gas are unevenly spread throughout its weak spiral arms, and its core is just offset from the midpoint of its extremes. Its arms are also asymmetrical, thought to have been influenced by the gravitational pull of other galaxies within its group.

M105 is an elliptical galaxy 36.6 million light-years away. It is the biggest elliptical galaxy in the Messier catalogue that is not in the *Virgo cluster*. The galaxy also contains a few young stars and stellar clusters, suggesting some elliptical galaxies still form new stars, but very slowly. It lies near another *elliptical galaxy NGC 3371* (Mag 9.9), and may be seen in the same frame.

NGC 2903 (Spiral Galaxy) Medium (Mag 9.00)

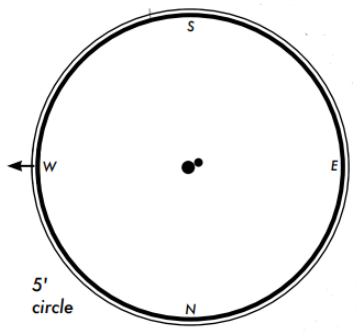
RA 9h 32m 10s | Dec +21° 30′ 03″

NGC 2903 is an isolated barred spiral galaxy, and one of the brighter galaxies visible from the Northern Hemisphere, in the constellation of *Leo*. This galaxy is located about 30 million light-years away from the Milky Way, and is a member of the *Virgo Supercluster*.

Algieba (Multiple Star) Easy (Mag 2.23, 3.64)

RA 10h 19m 58s | Dec +19° 50′ 29″

Algieba (Gamma Leonis), is a binary star in the constellation *Leo*, made up of two red giants. It can be split easily into the pair, with an angular separation of 4.4”



The primary star (K type giant) is golden yellow in color, while it is more difficult to assign a color to its companion (G type giant) – some people see it as orange. Algieba is located 125 light years from us, and its companion orbits in an elliptical path more than 300 AU in diameter. It takes roughly 500 years to complete an orbit. It is currently close to its maximum separation, as seen from Earth. Evidence also indicates the presence of a planet roughly ten times Jupiter’s mass with a 1.2 AU-radius orbit and a 429-day period, although this is certainly not visible through our telescopes. (or any telescope for that matter)

(Photo credit: “Turn Left at Orion” - Guy Consolmagno, Dan M. Davis)

Leo I Dwarf Galaxy Insane or Astrophotography (Mag 10.10):

RA 10h 08m 27s | Dec +12° 18′ 27″

The Leo I Dwarf Galaxy is a dwarf spheroidal galaxy in the constellation *Leo*. At about only 820,000 light-years distant, it is a member of the *Local Group* of galaxies and is thought to be one of the most distant satellites of the Milky Way galaxy. It is located only 12 arc minutes from *Regulus*, the brightest star in the constellation. The proximity of *Regulus* and the low surface brightness make it a real challenge to observe, making it deserve an Insane difficulty rating.

Sextans and Hydra

NGC 3242 - Ghost of Jupiter Nebula (Planetary Nebula) Medium (Mag 7.30)

RA 10h 24m 46s | Dec −18° 38′ 32″

The Ghost of Jupiter Nebula is a planetary nebula located in the constellation *Hydra*. The nebula measures around two light years long from end to end, and contains a central white dwarf with an apparent magnitude of 11. The inner layers of the nebula were formed some 1,500 years ago. It can easily be observed with amateur telescopes and appears bluish-green to most observers with larger telescopes distinguishing the outer halo as well.

NGC 3115 - Sextans Spindle Galaxy Medium (Mag 8.89)

RA 10h 05m 14s | Dec −7° 43′ 07″

The Spindle Galaxy is a bright, field (isolated), lenticular (S0) galaxy in the constellation *Sextans*. A lenticular galaxy is of the form in between a spiral and an elliptical galaxy. At about 32 million light-years away from Earth, it is several times bigger than the Milky Way. It is a lenticular (S0) galaxy because it contains a disk and a central bulge of stars, but without a detectable spiral pattern. NGC 3115 is seen almost exactly edge-on. It has very little gas and dust left that would trigger new star formation. The vast majority of its component stars are very old.

Coma Berenices

M3 (Globular Cluster) Easy (Mag 6.30)

RA 13h 42m 12s | Dec +28° 22′ 38″

M3 is a globular cluster located 33.9 thousand light years from Earth in the northern constellation of *Canes Venatici*, close to the border with *Coma Berenices*. It is one of the largest and brightest globular clusters discovered with around 500,000 stars, and one of the finest in the sky. It is estimated to be 11.4 billion years old, and centered at 32,600 light-years away from Earth. It is quite isolated as it is 31,600 light-years above the Galactic plane and roughly 38,800 light-years from the center of the Milky Way.

M53 (Globular Cluster) Medium (Mag 7.69)

RA 13h 12m 55s | Dec +18° 10′ 05″

M53 is a globular cluster in the *Coma Berenices* constellation. It is one of the more outlying globular clusters, being about 60,000 light-years light-years away from the Galactic Center, and almost the same distance (about 58,000 light-years) from the Solar System.

This is considered a metal-poor cluster and at one time was thought to be the most metal-poor cluster in the Milky Way. Abundance measurements of cluster members on the red giant branch show that most are first-generation stars. That is, they did not form from gas recycled from previous generations of stars in the cluster. This differs from the majority of globular clusters that are more dominated by second generation stars. The second generation stars in M53 tend to be more concentrated in the core region.

M64 - Black Eye Galaxy Medium (Mag 8.50)

RA 12h 56m 44s | Dec +21° 40′ 58″

The Black Eye Galaxy is a relatively isolated spiral galaxy 17 million light-years away in the northern constellation of *Coma Berenices*, and is a part of the *Virgo Supercluster*. A dark band of absorbing dust partially in front of its bright nucleus gave rise to its nicknames of the "Black Eye", “Evil Eye” or “Sleeping Beauty”.

The interstellar medium of Messier 64 consists of two counter-rotating disks that are approximately equal in mass. The inner disk contains the prominent dust lanes of the galaxy. The stellar population of the galaxy exhibits no measurable counter-rotation. Possible

formation scenarios include a merger with a gas-rich satellite galaxy in a retrograde orbit, or the continued accretion of gas clouds from the intergalactic medium. It has a diameter of 53,800 light-years.

M99 - Coma Pinwheel Galaxy Medium (Mag 9.89)

RA 12h 18m 50s | Dec +14° 24′ 59″

The Coma Pinwheel Galaxy is a grand-design face-on spiral galaxy in the northern constellation *Coma Berenices* approximately 49 million light-years from the Milky Way.This galaxy has a morphological classification of SA(s)c, indicating a pure spiral shape with loosely wound arms. It has a peculiar shape with one normal looking arm and an extended arm that is less tightly wound.

A bridge of neutral hydrogen gas links NGC 4254 with VIRGOHI21, an HI region and a possible dark galaxy - a galaxy made nearly purely of dark matter and without stars, which can be detected only due to its gas presence. The gravity from the latter may have distorted M99 and drawn out the gas bridge, as the two galaxy-sized objects may have had a close encounter before parting greatly. However, VIRGOHI21 may instead be tidal debris from an interaction with the lenticular galaxy NGC 4262 some 280 million years ago.

M100 - Mirror Galaxy Hard (Mag 9.39)

RA 12h 22m 55s | Dec +15° 49′ 21″

The Mirror Galaxy is a grand-design intermediate spiral galaxy in the southern part of the northern constellation *Coma Berenices*. It is one of the brightest and largest galaxies in the Virgo Cluster and is approximately 55 million light-years from our galaxy, about 166,000 light-years in diameter.

Messier 100 is considered a starburst galaxy with the strongest star formation activity concentrated in its center, within a ring – actually two tightly wound spiral arms attached to a small nuclear bar of radius: one thousand parsecs – where star formation has been taking place for at least 500 million years in separate bursts.

NGC 4565 - Needle Galaxy Medium (Mag 9.60)

RA 12h 36m 21s | Dec +25° 59′ 16″

The Needle Galaxy is an edge-on spiral galaxy about 30 to 50 million light-years away in the constellation *Coma Berenices*. It is known as the Needle Galaxy for its narrow profile, and is a very prominent example of its class.

Ursa Major

M81 & M82 - Bode & Cigar Galaxies: Medium (Mag 6.90, 8.39)

RA 9h 55m 33s | Dec +69° 3′ 55″ (M81)

M81, Bode Galaxy, is a grand-design spiral galaxy about 12 million light-years away in the constellation *Ursa Major*. It has a diameter of 96,000 light-years. With high surface brightness, it can be seen prominently.

M82, Cigar Galaxy, is a starburst galaxy approximately 12 million light-years away in the constellation *Ursa Major*. It has a high surface brightness, but appears edge on.
It is the second-largest member of the M81 Group, after M81, with a diameter of 40,800 light-years. It is about five times more luminous than the Milky Way and its central region is about one hundred times more luminous. In November 2023, a gamma-ray burst was observed in M82, which was determined to have come from a magnetar, the first such event detected outside the Milky Way.

Both of these galaxies are north of +60°, so cannot be seen with the Northumberland, but instead can be seen with the Thorrowgood or Dobsonian telescopes.

M97 - Owl Nebula (Planetary Nebula) Hard (Mag 9.80)

RA 11h 14m 48s | Dec +55° 1′ 0″
The Owl Nebula is a planetary nebula approximately 2,030 light years away in the constellation *Ursa Major*. Estimated to be about 8,000 years old, it is approximately circular in cross-section with a faint internal structure.
It was formed from the outflow of material from the stellar wind of the central star as it evolved along the asymptotic giant branch. The nebula is arranged in three concentric shells, with the outermost shell being about 20–30% larger than the inner shell. The owl-like appearance of the nebula is the result of an inner shell that is not circularly symmetric, but instead forms a barrel-like structure aligned at an angle of 45° to the line of sight. The 14th magnitude central star has passed the turning point in its evolution and is condensing to form a white dwarf.

M101 - Pinwheel Galaxy Hard (Mag 7.90)

RA 14h 3m 13s | Dec +54° 20′ 57″
The Pinwheel Galaxy is a face-on, counterclockwise intermediate spiral galaxy located 21 million light-years from Earth in the constellation *Ursa Major*. Photography is recommended to make out the detail on the spiral arms.

M51 - Whirlpool Galaxy Medium (Mag 8.39 & 9.60)

RA 13h 29m 53s | Dec +47° 11′ 43″
The Whirlpool Galaxy is an interacting grand-design spiral galaxy with a Seyfert 2 active galactic nucleus, in the constellation *Canes Venatici*, near the border with *Ursa Major*. It is 31 million light-years away and 76,900 light-years in diameter.
It can be observed distinctly due to a high surface brightness, and has a companion *dwarf galaxy NGC 5195* of Mag 9.60, with which it is interacting, and the companion may be made out in good conditions.

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